

AMENDMENT NO. 2 FEBRUARY 2024
TO
IS 800 : 2007 GENERAL CONSTRUCTION IN STEEL — CODE OF PRACTICE
(Third Revision)

(Page 12, clause 2.2.2) — Insert the following at the end:

‘Steel tubes used in building construction shall be hot finished tubes conforming to the requirements specified in IS 1161.’

(Page 12, clause 2.2.3) — Insert the following at the end:

‘Tubes made by other than hot finishing processes, or which have been subjected to cold working, shall be regarded as hot finished if they have subsequently been heat-treated and are supplied in the normalized condition.

NOTE — Grade ERW tubes specified in IS 1161 with a carbon content less than 0.30 percent, may be considered as hot finished for the purpose of 2.2.3.’

[Page 13, Table 1, Sl No. (iv), col (3) and (4)] — Substitute the following for existing:

Grade/Classification	Yield Stress MPa, <i>Min</i>
(3)	(4)
4.6	240
4.8	320
5.6	300
5.8	400
6.8	480

[Page 14, Table 1, Sl No. (viii)] — Substitute the following for existing:

Sl No.	Indian Standard	Grade/ Classification	Properties				
			Yield Stress MPa, <i>Min</i>	Ultimate Tensile Stress MPa, <i>Min</i>	Elongation Percent, <i>Min</i>		
(1)	(2)	(3)	(4)	(5)	(6)		
<i>d or t (in mm)</i>							
			< 20	20 to 40	> 40		
viii)	IS 2062	E250 (A, BR, B0, C)	250	240	230	410	23
		E275 (A, BR, B0, C)	275	265	255	430	22
		E300 (A, BR, B0, C)	300	290	280	440	22
		E350 (A, BR, B0, C)	350	330	320	490	22
		E410 (A, BR, B0, C)	410	390	380	540	20
		E450 (A, BR)	450	430	420	570	20

Price Group 4

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(Page 16, clause 3.3, line 8) — Substitute ‘Imposed Load’ for ‘Live Load’.

[Page 17, clause 3.7.2 (d)] — Insert the following at the end of the para:

‘The design of slender web elements in flexural members may be done as given in 8.2.1.1 for flexural and 8.4.2.2 for shear.’

[Page 18, Table 2, row 2, col (3) and (4)] — Substitute ‘ 42ε ’ for “‘Not applicable’ for entry against ‘Axial compression’”.

[Page 18, Table 2, row 3, col (3) and (4)] — Substitute ‘ 42ε ’ for “‘Not applicable’ for entry against ‘Axial compression’ of ‘Web of an I, H or box section’”.

[Page 18, Table 2, row 6, col (3) and (4)] — Substitute the following for the existing for entries against single angle, or double angles with the components separated, axial compression (all three criteria should be satisfied)

Ratio	Class of Section	
	Class 1 Plastic	Class 2 Compact
(2)	(3)	(4)
b/t		15.7ε
d/t		15.7ε
$(b+d)/t$		25ε

(Page 21, clause 3.10.4) — Substitute the following for existing:

‘When the provisions of these sections are met for a building or open structure made only of steel frame with elastic steel claddings, the stress analysis due to temperature is not required. However, for buildings with steel concrete composite constructions, specialist literature may be referred to decide on thermal stress design.’

(Page 21, Fig. 4, Title) — Insert ‘COVERED’ between ‘OF’ and ‘BUILDING’.

[(Page 27, clause 5.1.2, para 2, Sl No. (a), line 25)] — Substitute ‘imposed load’ for ‘live load’.

(Page 29, Table 4) — Substitute ‘IL’ for ‘LL’, wherever appearing.

[Page 30, clause 5.5.1.1 (b), line 3] — Substitute ‘Partial safety factor for Loads’ for ‘Load factors’.

(Page 31, clause 5.6.1, line 6 and 7) — Substitute ‘partial safety factor in Table 4 for loads, as required in Table 6’ for ‘a load factor of 1.0’.

(Page 31, Table 6) — Substitute ‘Imposed Load’ for ‘Live load’, wherever appearing.

(Page 32, clause 5.6.2, line 2) — Substitute ‘imposed load’ for ‘live load’.

(Page 46, clause 7.4.1, para 2) — Substitute the following for the existing:

‘The nominal bearing pressure between the base plate and the support below may be determined on the basis of a linearly varying distribution of pressure. The maximum bearing pressure on concrete under the limit state design load should not exceed the bearing strength equal to $0.45 f_{ck} [\sqrt{A_1/A_2}]$ and $0.9 f_{ck}$, whichever is smaller, where f_{ck} is the smaller of characteristics cube strength of concrete or bedding material, A_1 is the area of concrete pedestal on which the base plate is sitting and A_2 is the surface area of the steel base plate transferring the load to the concrete pedestal.’

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(Page 47, clause **7.5.1**, line 2) — Substitute ‘connection’ for ‘gusset’.

(Page 48, clause **7.5.1.2**, see also Amendment No. 1) — Substitute the following with the existing:

‘The design strength of single angle members loaded in compression through one of its legs is affected by flexural torsional buckling and bending. The design compressive strength of such eccentricity loaded angle members, *in-lieu* of a more exact second order design under combined bending axial compression, may be evaluated as given below.

The design compressive strength, of single angles loaded through gusset connection to one leg parallel to the *a-a* axis (*a-a* axis being either the 2-2 or the *y-y* axis, depending on which leg is connected to the gusset:

$$f_{cde} = K_f \chi_{aa} f_y / \gamma_{m0}$$

where

$$K_f = k_1 + k_2 \lambda_{aa} + k_3 \lambda_\phi ; \text{ and}$$

χ_{aa} = stress reduction factor [see Table 8 (b)] for buckling class ‘b’, calculated as per **7.1.2.1**, using the non-dimensional effective slenderness ratio, λ_{aa} .

where

K_f = modification factor to account for eccentric end connection; and
 k_1, k_2, k_3 = constants depending upon the end condition, as given in Table 12.

$$\lambda_{aa} = \frac{\left(\frac{l_{aa}}{r_{aa}}\right)}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}}$$

$$\lambda_\phi = \frac{(b_1 + b_2) / 2t}{\varepsilon \sqrt{\frac{\pi^2 E}{250}}}$$

l_{aa} = centre-to-centre length of the lateral support preventing translation of the member perpendicular to *a-a* axis (*a-a* axis being either the *z-z* or the *y-y* axis, depending on which leg is connected to the gusset);

r_{aa} = radius of gyration of the angle member about the *a-a* axis (*a-a* axis being either the *z-z* or the *y-y* axis, depending on which leg is connected to the gusset), parallel to the plane of the end gusset plates;

ε = is the yield stress ratio, $\sqrt{\frac{250}{f_y}}$, f_y being in MPa, ;

b_1, b_2 = width of the connected and outstanding legs of the angle, respectively; and

t = thickness of the leg. ’

(Page 48, Table 12) — Substitute the following for the existing:

SI No.	End Connection	Gusset/Connecting Member Fixity ¹⁾	k_1	k_2	k_3
(1)	(2)	(3)	(4)	(5)	(6)
i)	Fully welded or connected with two or more bolts	Fixed	0.798	0.563	-2.072
		Hinged	0.401	0.420	-1.040
ii)	Single bolt	Fixed	0.418	0.547	-1.400
		Hinged	0.374	0.415	-2.072

¹⁾ In-plane rotational restraint provided to the gusset by the supporting member. For partial restraint, K_f can be interpolated between the results for fixed and hinged cases.

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[Page 49, Fig. 10, title, (see also Amendment No. 1)] — Substitute ‘LACED COLUMN LACING PATTERNS’ for ‘TOP RESTRAINT CONDITIONS’.

(Page 54, clause **8.2.2.1**, last para) — Insert the following line at the end:

‘The above equations for calculating M_{cr} may be conservatively used for channel sections.’

(Page 58, Table 15) — Add ‘only’ before compression, wherever appearing.

(Page 58, Table 15) — Substitute ‘no warping restraint’ for ‘warping not restrained’, wherever appearing.

(Page 58, Table 15, Note 3) — Insert the following after Note 3:

‘4 In case of continuous beams, L shall be taken as the distance between points of inflection, and the restraint conditions at the points of inflection should be considered.’

[Page 60, clause **8.4.2.2 (b)**, description of ‘c’] – Substitute ‘spacing of transverse stiffeners in the web’ for ‘spacing of stiffeners in the web’.

[Page 60, clause **8.5.1**, para 1, (see also Amendment No. 1)] – Substitute the following for the existing:

‘The girders in which the interior panel (Panel A) is designed using tension field action, the end panel (Panel B) shall be designed using only simple post-critical method, according to **8.4.2.2 (a)**’.

(Page 61, Table 16, row 5) — Substitute ‘conditions’ for ‘consitions’.

(Page 62, clause **8.5.2**, para 1) — Substitute the following for the existing:

‘The end panel (Panel A) shall be designed according to **8.4.2.2 (b)**.’

[Page 66, clause **8.7.1.5**, para 3, SI No. (a), line 1] — Insert ‘,’ between ‘ $0.7L$ ’ and ‘when’.

[Page 68, clause **8.7.9 (b)**, description of α_s , equation 2] — Substitute ‘ ≤ 100 ’ for ‘ $= 100$ ’.

(Page 69, clause **8.9**, first sentence) — Substitute the following with the existing first sentence:

‘All purlin shall be designed in accordance with the requirements for beams as specified in **8.2.1** and **8.2.2**. This limitations of bending stress based on lateral instability of the compression flange shall be considered if the purlin does not receive adequate support from bracings and sheathings in the plane of compression flange.’

(Page 73, clause **10.2.1**, line 2) — Substitute ‘, as follows:’ for ‘.’.

(Page 76, clause **10.4.2**, first sentence) — Substitute the following for the existing:

‘Where slip between bolted plates cannot be tolerated at working loads/strength limit state loads (slip critical connections), the requirements of **10.4.3** shall be satisfied at service and strength limit states loads, respectively. However, the requirements of **10.4.4** shall be satisfied by all connections at the strength limit states.’

(Page 77, clause **10.4.7**, line 2) — Delete ‘is significant, it’.

(Page 84, clause **11.1.3**, lines 2 and 3) — Substitute ‘partial safety factors for loads, specified in Table 4 for the limit states of serviceability.’ for ‘partial load factor for service load/working load.’

(Page 84, clause **11.3.1**, description of f_{cd}) — Insert the following at the end of sentence:

‘,calculated without the partial safety factor, γ_{m0} .’

[Page 85, clause **11.4.1 (c)**, description of f_y] — Substitute ‘Material’ for ‘Sect’.

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(Page 87, clause **12.1**, first sentence) — Insert the following at the end:

‘Ductility requirements in **12.4** to **12.12** need not be satisfied in case of non-ductile frames as defined in **12.13**.’

[Page 87, clause **12.2.3 (a)**, line 1] — Substitute ‘Imposed Load’ for ‘Live Load’.

[Page 87, clause **12.2.3 (b)**] — Insert the following at the end:

‘NOTE — The above combinations need to be considered for only the design axial forces of members, ignoring simultaneous bending moments.’

(Page 87, Table 23) — Substitute the following for the existing:

Table 23 Response Reduction Factor, R, for Building System

(Clause 12.3)

Sl No.	Lateral Load Resisting System	R
(1)	(2)	(3)
i)	Braced frame systems:	
	a) Ordinary concentrically braced frame (OCBF)	4.0
	b) Special concentrically braced frame (SCBF)	4.5
	c) Ordinary eccentrically braced frame (OCBF)	4.5
	d) Special concentrically braced frame (OCBF)	5.0
ii)	Moment frame system:	
	a) Ordinary moment resisting frame	4.0
	b) Special moment resisting frame	5.0
iii)	Non-ductile frame	1.5

(Page 87, clause **12.3**) — Insert new para at the end:

‘The braced and moment resisting frames in single storey industrial buildings can be designed as non-ductile frames, if their overall height to overall width ratio does not exceed 1.0. Such frames need not meet the ductility requirements of other types of frames as specified in **12.4** to **12.12**.’

(Page 87, clause **12.4.1**, line 1) — Insert ‘critical zones of’ between ‘in’ and ‘frames’.

(Page 87, clause **12.4.1**, para 1) — Insert the following at the end:

‘Critical zones of frames are zones of plastic hinges of moment resisting frames and beam to columns as well as bracing and their connections of braced frames. The plastic hinge zones shall be taken as a distance equal to 2 times the depth of the member from the section of maximum bending moment.’

(Page 87, clause **12.4.2**, line 1) — Substitute ‘welded connections used in the critical zones of’ for ‘welds used in’.

(Page 88, clause **12.5.1**, description of P_r) — Substitute the following for the existing:

‘ P_r = required compressive strength of the member as evaluated for load combination in Table 4; and’

(Page 88, clause **12.5.1**, description of P_d) — Substitute ‘strength’ for ‘stress’.

(Page 88, clause **12.5.1.1**) — Insert the following at the end:

‘without considering the simultaneous moments as obtained from the analysis.’

(Page 88, clause **12.5.1.2**, line 3) — Substitute ‘axially loaded member’ for ‘column’.

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(Page 88, clause **12.5.2.2**, line 2) — Substitute ‘shown in’ for ‘showing’.

(Page 88, clause **12.5.2.2**, line 3 and 4) — Substitute ‘where A_f is the area of each flange in the connected column, and 1.2 times $F_y A_w$ as minimum required strength in cases where web splices are provided.’ for ‘where A_f is the area of each flange in the smaller connected column.’

(Page 88, clause **12.7.2.3**, line 1) — Insert ‘at all levels’ after ‘braces’.

(Page 88, clause **12.7.2.3**, line 4) — Substitute ‘floor shear at the level’ for ‘lateral load.’

(Page 88, clause **12.7.2.5**, last line) — Insert the following at the end:

‘in such built-up braces.’

[Page 88, clause **12.7.3.1 (c)**] — Insert the following at the end:

‘, based on analysis.’

(Page 89, clause **12.8.2.2**, last line) — Substitute ‘(except for hanger elements, which support only gravity load)’ for ‘(only hangers)’.

(Page 89, clause **12.10**, title) — Substitute the following for the existing:

‘Ordinary Moment Resisting Frames (OMRF)’

(Page 89, clause **12.10.1**, line 1) — Substitute ‘Ordinary moment resisting frames (OMRF)’ for ‘Ordinary Moment Frames (OMF)’.

(Page 89, clause **12.10.1**, line 5) — Insert ‘resisting’ between ‘moment’ and ‘frames’.

(Page 89, clause **12.10.1.1**, line 1) — Substitute ‘Ordinary moment resisting frames (OMRF)’ for ‘Ordinary moment frames (OMF)’.

(Page 90, clause **12.10.2.4**, line 3) — Substitute ‘IL’ for ‘LL’.

(Page 90, clause **12.11.1**, Title) — Substitute the following for the existing:

‘Special Moment Resisting Frames (SMRF)’.

(Page 90, clause **12.11.1**, line 1) — Substitute ‘Special moment resisting frames (SMRF)’ for ‘Special moment frames (SMF)’.

(Page 90, clause **12.11.1**, line 6) — Substitute ‘Special moment resisting frames (SMRF)’ for ‘Special moment frames (SMF)’.

(Page 90, clause **12.11.1.1**, line 1) — Substitute ‘Special moment resisting frames (SMRF)’ for ‘Special moment frames (SMF)’.

(Page 91, clause **12.12.2**) — Insert the following at the end:

‘12.13 Non-Ductile Braced and Moment Resisting Frames

Frames which do not meet the ductility requirements of the **12.4** to **12.12** and designed as per the response reduction factor for non-ductile frames specified in Table 23 shall be classified as non-ductile frames. These frames shall have height to span ratio not exceeding 1.0, such as in industrial buildings.’

[Page 91, clause **13.2.1 (h)**] — Substitute ‘Unfilled holes’ for ‘Holes’.

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[Page 91, clause **13.2.1**, para 2] — Substitute the following for the existing:

‘Fatigue need not be investigated if condition in **13.6 (b)** is satisfied.

[Page 91, clause **13.2.1**, para 3] — Insert the following as title for para 3 and para 4:

‘**13.2.1.1 Thickness correction factor**’

(Page 92, clause **13.2.2.3**) — Delete.

(Page 99, clause **13.5.1**) — Delete.

(Page 99, clause **13.5.2.4**, line 2) — Substitute ‘ f_i or τ_i ’ for ‘ f_{fi} or τ_{fi} ’.

[Page 100, clause **13.5.2.4 (a)**, formula] — Substitute ‘ r_5 ’ for ‘ γ_5 ’ and ‘ μ_r ’ for ‘ μ ’.

[Page 100, clause **13.5.2.4 (b)**, formula] — Substitute ‘ r_5 ’ for ‘ r ’; ‘ τ_i^5 ’ for ‘ τ_{fi}^5 ’; and ‘ μ_r ’ for ‘ μ ’.

[Page 100, clause **13.5.2.4 (b)**] — Substitute ‘ r_5 ’ for ‘ γ_5 ’ and ‘ μ_r ’ for ‘ μ_c ’, wherever appearing.

[Page 100, clause **13.5.2.4 (b)**, line 7] — Insert ‘shear stress ranges,’ before ‘ τ_i ’.

[Page 100, clause **13.6 (b) (1)**, formula] — Substitute ‘ μ_r ’ for ‘ μ_c ’.

[Page 100, clause **13.6 (b) (2)**, formula] — Substitute ‘ μ_r ’ for ‘ μ_c ’.

[Page 100, clause **13.6 (b) (3)**] — Substitute the following for the existing:

3) The total number of actual stress cycles N_{sc} , satisfies:

$$N_{sc} \leq 5 \times 10^6 \left(\frac{27 \mu_r}{\gamma_{mft} f_{feq}} \right)^3$$

where

f_{feq} = equivalent constant amplitude stress range, in MPa, given by:

$$f_{feq} = \left[\frac{\sum_{i=1}^{r_5} n_i f_i^3 + \sum_{j=r_5}^r n_j f_j^5}{n} \right]^{1/3}$$

where

$$n = \sum_{i=1}^r n_i$$

f_i, f_j = stress ranges falling above and below the f_{fn} , the stress range corresponding to the detail at 5×10^6 number of life cycles.

r_5 = summation upper limit of all the normal stress ranges (f_i) having magnitude lesser than $(\mu_r f_{fn} / \gamma_{mft})$ for that detail and the lower limit of all the normal stress ranges (f_j) having magnitude greater than $(\mu_r f_{fn} / \gamma_{mft})$ for the detail.

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(Page 105, clause **16.1**, para 1) — Insert the following at the end:

‘The provisions of this section are quite rudimentary and simplistic. Specialist literature for design of steel structures for fire resistance and design based on testing may be necessary for a more comprehensive design of steel buildings to resist fire.’

[Page 106, clause **16.3.2 (a) (2)**, line 1] — Insert ‘(t)’ after ‘time’.

[Page 106, clause **16.3.2 (a) (2)**, line 3] — Delete ‘(t)’.

(Page 109, clause **16.11**, last line) — Insert ‘see’ before ‘Table 31’.

(Page 110, clause **17.2.2.1**) — Substitute the following for the existing:

‘The holes for bolts may be made as given in Table 19, unless otherwise specified by the engineer. The hole diameter in base plate shall not exceed the anchor bolt diameter by more than 6 mm.’

(Page 134, Table 44, description of ‘g’) — Insert ‘in mm’ at the end of the line.